Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 02: Plane Waves

Instructor: Zihan Wang

Monochromatic plane wave

$$\vec{E} = \vec{E_0} e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

$$\vec{B} = \vec{B_0} e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

is one solution to Maxwell Eq.

Property of Monochromatic plane wave

- Divergence

$$\vec{E} = \vec{E_0} e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

$$\vec{B} = \vec{B_0} e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

$$\Rightarrow \begin{cases} \nabla \cdot \vec{E} = 0 \\ \nabla \cdot \vec{B} = 0 \end{cases} \Rightarrow \quad i\vec{k}\cdot\vec{E} = 0, \quad i\vec{k}\cdot\vec{B} = 0$$

$$\Rightarrow \quad \vec{k} \perp \vec{E}, \quad \vec{k} \perp \vec{B}$$

$$\Rightarrow \quad \text{Transverse wave}$$

Property of Monochromatic plane wave

- Curl

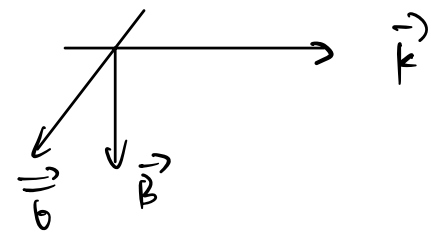
$$\vec{E} = \vec{E_0} e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

$$\vec{B} = \vec{B_0} e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

$$\Rightarrow \begin{cases} \nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{B} = \mu_0 \varepsilon_0 \dfrac{\partial \vec{E}}{\partial t} \end{cases}$$

$$\Rightarrow \quad i\vec{k} \times \vec{E} = -(-i\omega \vec{B}) = i\omega \vec{B}$$

$$\Rightarrow \quad \vec{k} \times \vec{E} = \vec{B}\omega$$



$$k|\vec{E}| = \omega|\vec{B}| \quad \Rightarrow \quad |\vec{E}| = \frac{\omega}{k}|\vec{B}| = c|\vec{B}|$$

Property of Monochromatic plane wave

- Phase velocity

$$\vec{E} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

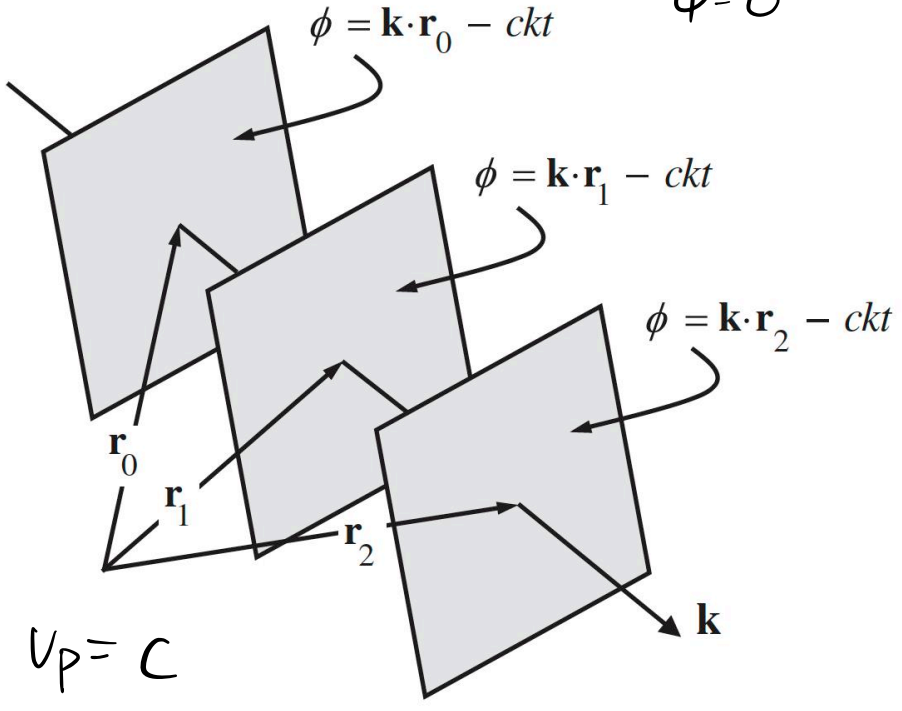
$$\phi = \vec{k}\cdot\vec{x} - \omega t \qquad \text{Phase}$$

$$= k\left(\hat{k}\cdot\vec{x} - \frac{\omega}{k}t\right)$$

$$= k(\hat{k}\cdot\vec{x} - v_p t)$$

$$v_p = \frac{\omega}{k} \qquad \text{In vacuum, } v_p = c$$

Track a surface
$\phi = 0$



Property of Monochromatic plane wave

- Impedance

$$\vec{H} = \frac{\vec{B}}{\mu_0} = \frac{\vec{k} \times \vec{b}}{\omega \mu_0} = \frac{\hat{k} \times \vec{b}}{\frac{\omega}{k} \mu_0}$$

$$\frac{|\vec{H}|}{|\vec{b}|} = \frac{1}{\frac{\omega}{k} \mu_0} = \frac{1}{\sqrt{\frac{1}{\mu_0 \varepsilon_0}}\, \mu_0} = \frac{1}{\sqrt{\mu_0 / \varepsilon_0}}$$

$$Z_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} \quad \Rightarrow \quad |\vec{H}| = \frac{|\vec{b}|}{Z_0}$$

Property of Monochromatic plane wave

- Electromagnetic energy density

$$u(\vec{x},t) = \frac{1}{2}\varepsilon_0\left(|\mathrm{Re}\,\vec{E}|^2 + c^2|\mathrm{Re}\,\vec{B}|^2\right)$$

Unit: J/m^3

$$\langle u\rangle(\vec{x}) = \frac{1}{T}\int_0^T u\,dt \qquad T = \frac{2\pi}{\omega}$$

$$\mathrm{Re}\,\vec{E} = \frac{1}{2}\left(\vec{E} + \vec{E}^*\right)$$

$$u = \frac{1}{8}\varepsilon_0\left(\vec{E}^2 + \vec{E}^{*2} + 2\vec{E}\cdot\vec{E}^* + c^2\vec{B}^2 + c^2B^{*2} + 2c^2\vec{B}\cdot\vec{B}^*\right)$$

$$\langle u\rangle = \frac{1}{4}\varepsilon_0\left(\vec{E}\cdot\vec{E}^* + c^2\vec{B}\cdot\vec{B}^*\right) = \frac{1}{2}\varepsilon_0\,\vec{E}\cdot\vec{E}^*$$

Property of Monochromatic plane wave

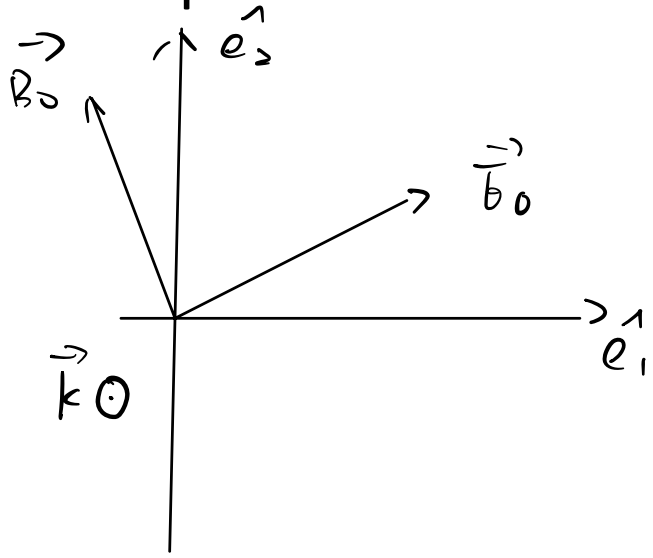
- Poynting Vector / Energy flux density $\vec{S} = \vec{E} \times \vec{H}$ W/m^2
- Momentum density
- Intensity $I = \langle u \rangle c = |\langle \vec{S} \rangle|$

$$\langle \vec{S} \rangle = \langle u \rangle c \hat{k}$$

$$\vec{g} = \varepsilon_0 \vec{E} \times \vec{B} \quad \Rightarrow \quad \vec{S} = c^2 \vec{g} \quad \Rightarrow \quad \langle \vec{g} \rangle = \frac{\langle u \rangle}{c} \hat{k}$$

Polarization of monochromatic plane wave

- Polarization Ellipse
- Real unit vectors $\hat{e}1$ and $\hat{e}2$
- $(\hat{e}1, \hat{e}2, \hat{k})$ is a right-handed orthogonal triad of unit vectors.



Polarization Ellipse

- The most general solution

$$\vec{E}_0 = \mathcal{E}_1 \hat{e}_1 + \mathcal{E}_2 \hat{e}_2$$

- $\mathcal{E}_1$ and $\mathcal{E}_2$ are complex scalars — Complex amplitudes in $\hat{e}_1$ and $\hat{e}_2$.

$$\mathcal{E}_1 = A e^{i\delta_1} \qquad \mathcal{E}_2 = B e^{i\delta_2}$$

$$\vec{E}(\vec{x}, t) = A e^{i(\vec{k}\cdot\vec{x} - \omega t + \delta_1)} \hat{e}_1 + B e^{i(\vec{k}\cdot\vec{x} - \omega t + \delta_2)} \hat{e}_2$$

$$\mathrm{Re}\,\vec{E} = \underbrace{A\cos(\vec{k}\cdot\vec{x} - \omega t + \delta_1)\,\hat{e}_1}_{\vec{E}_1} + \underbrace{B\cos(\vec{k}\cdot\vec{x} - \omega t + \delta_2)\,\hat{e}_2}_{\vec{E}_2}$$

Polarization Ellipse

$$\frac{\bar{E}_2}{\bar{E}_1} = ?$$

$$\left(\frac{\bar{E}_1}{A}\right)^2 + \left(\frac{\bar{E}_2}{B}\right)^2 - 2\frac{\bar{E}_1}{A}\frac{\bar{E}_2}{B}\cos\delta = \sin^2\delta$$

$$\delta = \delta_2 - \delta_1$$

Linear polarization

$\delta = m\pi$

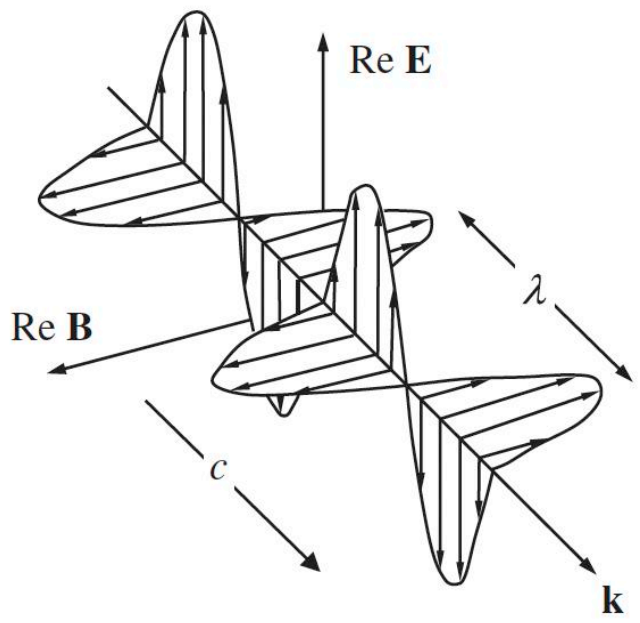
The polarization ellipse degenerates to a straight line Orthogonal electric field components are either in phase or 180◦ out of phase.

$$\vec{E}(\vec{x},t) = A e^{i(\vec{k}\cdot\vec{x}-\omega t+\delta_1)}\vec{e}_1 + B e^{i(\vec{k}\cdot\vec{x}-\omega t+\delta_2)}\vec{e}_2$$

$$\vec{E}(\vec{x},t) = A e^{i(\vec{k}\cdot\vec{x}-\omega t+\delta_1)}\vec{e}_1 \pm B e^{i(\vec{k}\cdot\vec{x}-\omega t+\delta_1)}\vec{e}_2$$

$$= (A\vec{e}_1 \pm B\vec{e}_2)\, e^{i(\vec{k}\cdot\vec{x}-\omega t+\delta_1)}$$

$$\vec{B} = \frac{\vec{E}}{c}$$



Circular polarization

$A = B$. $\delta = \frac{m\lambda}{2}$, m is odd.

$$\vec{U}(\vec{x}, t) = A e^{i(\vec{k}\cdot\vec{x} - \omega t + \delta_1)} \vec{e}_1 + B e^{i(\vec{k}\cdot\vec{x} - \omega t + \delta_2)} \vec{e}_2$$

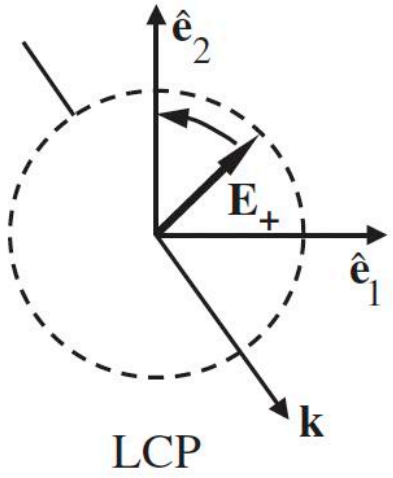
$$= A(\vec{e}_1 \pm i\vec{e}_2) e^{i(\vec{k}\cdot\vec{x} - \omega t + \delta_1)}$$

Polarization Ellipse

- Circular polarization

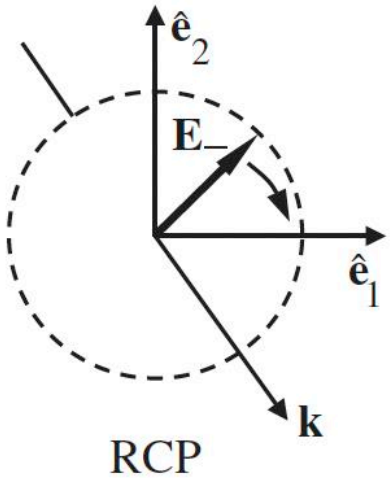
$\vec{E}_+ = A(\hat{e}_1 + i\hat{e}_2)e^{\cdots}$

$\vec{E}_- = A(\hat{e}_1 - i\hat{e}_2)e^{\cdots}$



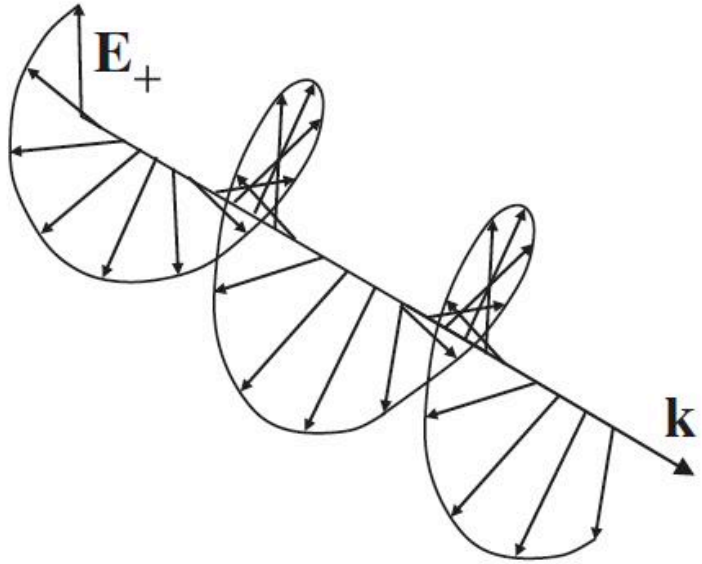
LCP

Left



RCP

right



Complex unit vectors

$$\hat{\mathbf{e}}_+ = \frac{1}{\sqrt{2}}(\hat{\mathbf{e}}_1 + i\,\hat{\mathbf{e}}_2) \qquad \hat{\mathbf{e}}_- = \frac{1}{\sqrt{2}}(\hat{\mathbf{e}}_1 - i\,\hat{\mathbf{e}}_2).$$

$$\mathbf{E}(\mathbf{r}, t) = [\mathcal{E}_+\hat{\mathbf{e}}_+ + \mathcal{E}_-\hat{\mathbf{e}}_-]\exp\{i(\mathbf{k}\cdot\mathbf{r} - \omega t)\}.$$

$= (\xi_1 \vec{e_1} + \xi_2 \vec{e_2}) \quad \cdots$

Plane waves in a nonconducting medium

- ϵ and μ are in general complex numbers depending on ω. Currently, we assume they are real and positive (no losses).

$$\varepsilon(\vec{x}, t, \omega) = \varepsilon_r + i\varepsilon_i$$

$$\Rightarrow \quad \varepsilon = \varepsilon_r$$

$$\begin{cases} \nabla \cdot \vec{D} = 0 & \vec{D} = \varepsilon \vec{E} \\ \nabla \cdot \vec{B} = 0 & \vec{B} = \mu \vec{H} \\ \nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} & \\ \nabla \times \vec{H} = \dfrac{\partial \vec{D}}{\partial t} & \end{cases}$$

Plane waves in a nonconducting medium

$$\begin{cases} \nabla^2 \vec{G} - \mu\epsilon \dfrac{\partial^2 \vec{G}}{\partial t^2} = 0 \\ \nabla^2 \vec{B} - \mu\epsilon \dfrac{\partial^2 \vec{B}}{\partial t^2} = 0 \end{cases}$$

$$v_p = \frac{1}{\sqrt{\mu\epsilon}}$$

$\vec{E} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x} - \omega t)}$ is still a solution.

Plane waves in a nonconducting medium

- Wave number

$$k = \omega\sqrt{\mu\varepsilon}$$

- Phase velocity

$$v_p = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\varepsilon}}$$

- Index of refraction

$$\eta = \frac{c}{v_p} = \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}}$$

- Impedance

$$Z = \sqrt{\frac{\mu}{\varepsilon}}$$

Example: Faraday Rotation

Faraday Rotation During Propagation For propagation along the z-axis, a medium supports left circular polarization with index of refraction n_L and right circular polarization with index of refraction n_R. If a plane wave propagating through this medium has $\mathbf{E}(z = 0, t) = \hat{\mathbf{x}} E \exp(-i\omega t)$, find the values of z where the wave is linearly polarized along the y-axis.